

Customer Retention Analysis Report

1. Project Overview

This project analyzes customer retention using a store dataset that consists of 30,801 customer records based on order behaviour, email engagement and service usage. The goal is to unfold the insights behind customer retention and identify the key factors that separate retained customers from non-retained customers.

2. Dataset Description

The dataset used in this project contains customer information related to orders, email activity, customer preferences, and retention. The original dataset contained 30,801 rows and 15 columns. The main target variable is **retained**, which shows whether a customer was retained or not.

Some important columns in the dataset include:

- **custid** – Customer identification number
- **created** – Date when the customer was created
- **firstorder** – Date of the customer's first order
- **lastorder** – Date of the customer's last order
- **retained** – Customer retention status
- **esent** – Number of emails sent
- **eopenrate** – Email open rate
- **eclickrate** – Email click rate
- **avgorder** – Average order value
- **ordfreq** – Order frequency
- **favday** – Customer's favorite day
- **city** – Customer's city
- **paperless, refill, doorstep** – Customer service usage information

The dataset contains both numerical and categorical variables. Date columns were converted into datetime format so they could be used for time-based analysis. After the data cleaning process, the dataset was reduced to 30,747 rows.

3. Exploratory Data Analysis (EDA):

The EDA in the Customer Retention Dataset is done using python and its libraries. Pandas is used for Data cleaning and matplotlib and seaborn is used for Data visualization. Only the key steps of the analysis are shown below.

1. Imported libraries, loaded the dataset and checked its dimensions

```
# Import libraries required for data analysis
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Load the Excel file
df = pd.read_excel('storedata.xlsx')

# Check the dataset dimensions
df.shape

(30801, 15)
```

2. Checked the dataset structure, summary statistics, missing values and duplicates

```
# Checked the dataset structure and summary statistics
df.info()
df.describe()

# Checked missing values and duplicate rows
df.isnull().sum()
df.duplicated().sum()

15 columns, 30801 entries
Missing values -> custid: 20, created: 20, firstorder: 20, lastorder: 20
Duplicate rows -> 0
```

3. Cleaned the data - removed missing values and converted date columns to datetime

```
# Data Cleaning
# Removed rows with missing customer / order date info
df = df.dropna(subset=['custid', 'created', 'firstorder', 'lastorder'])

# Converted firstorder and lastorder into datetime format
df['firstorder'] = pd.to_datetime(df['firstorder'], errors='coerce')
df['lastorder'] = pd.to_datetime(df['lastorder'], errors='coerce')
df = df.dropna(subset=['firstorder', 'lastorder'])

Cleaned dataset: 30747 rows, 15 columns, 0 missing values, 0 duplicates
```

4. Feature engineering - created service_score, estimated_emails_opened and customer_tenure

```
# Feature Engineering
# Created a service usage score
df['service_score'] = df['paperless'] + df['refill'] + df['doorstep']

# Estimated number of emails opened
df['estimated_emails_opened'] = df['esent'] * df['eopenrate']

# Derived customer tenure (in days) between first and last order
df['customer_tenure'] = (df['lastorder'] - df['firstorder']).dt.days
```

5. Outlier detection using the IQR method on avgorder, ordfreq and esent

```
# Outlier Detection using the IQR method
Q1 = df['avgorder'].quantile(0.25)
Q3 = df['avgorder'].quantile(0.75)
IQR = Q3 - Q1
lower, upper = Q1 - 1.5*IQR, Q3 + 1.5*IQR
outliers = df[(df['avgorder'] < lower) | (df['avgorder'] > upper)]

avgorder -> 1741 outliers
ordfreq -> 3723 outliers
esent -> 12 outliers
(All were kept since they may represent real customers.)
```

6. Calculated the overall customer retention rate

```
# Calculate customer retention rate
retention_rate = df['retained'].mean() * 100
print(retention_rate)

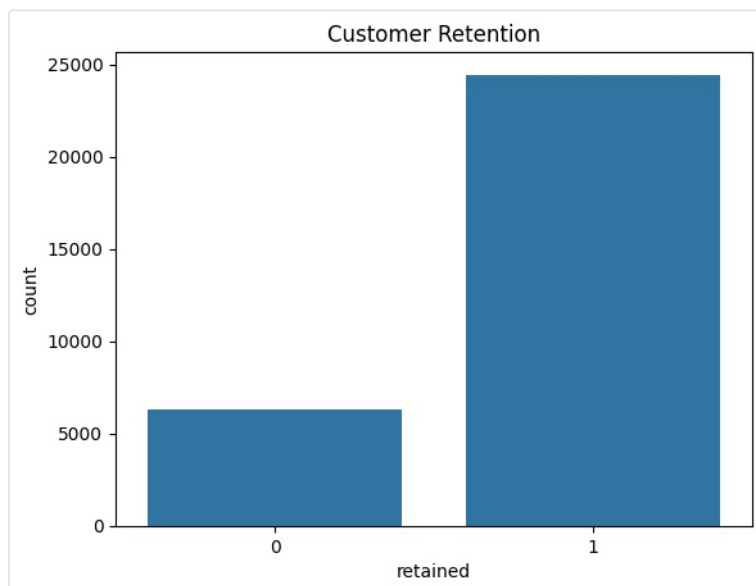
79.46466321917586
```

Overall Customer Retention Rate: 79.46% — calculated as the mean of the retained column across all cleaned records.

4. Data Visualization:

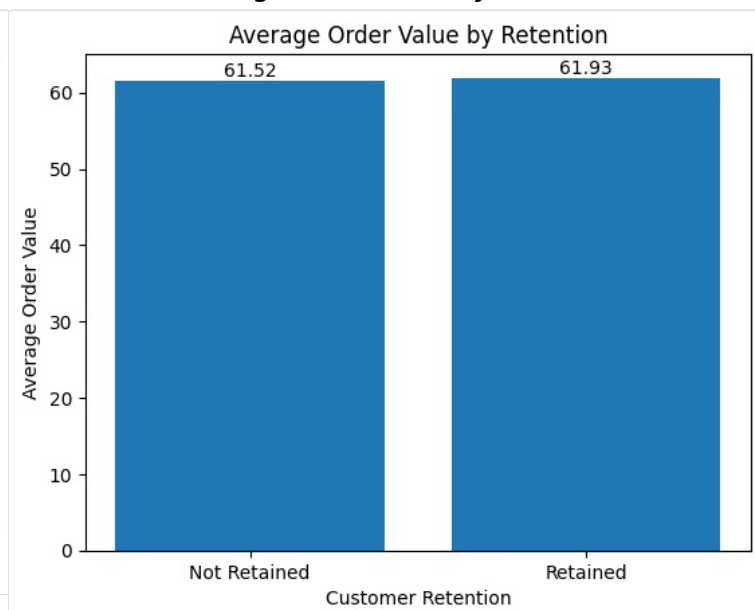
The data visualization is done using matplotlib and seaborn library in jupyter notebook after data cleaning using pandas.

1. Customer Retention distribution



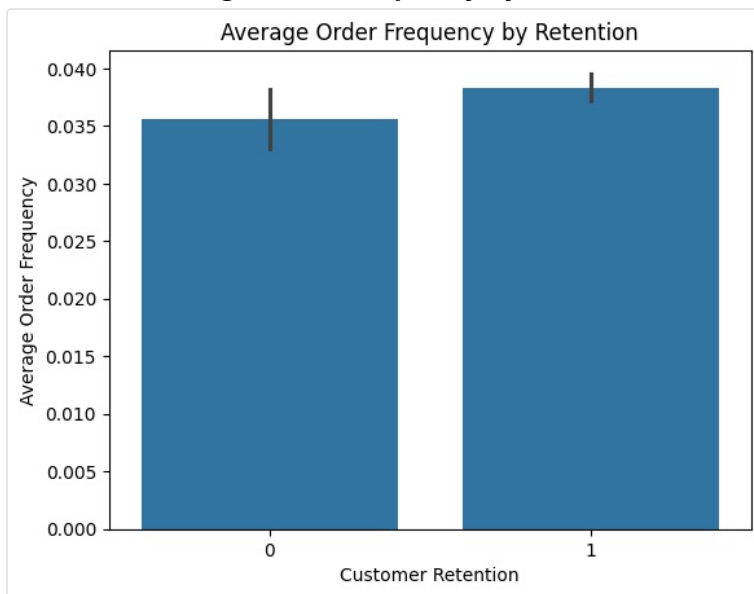
Most customers were retained, while fewer customers were not retained.

2. Average Order Value by Retention



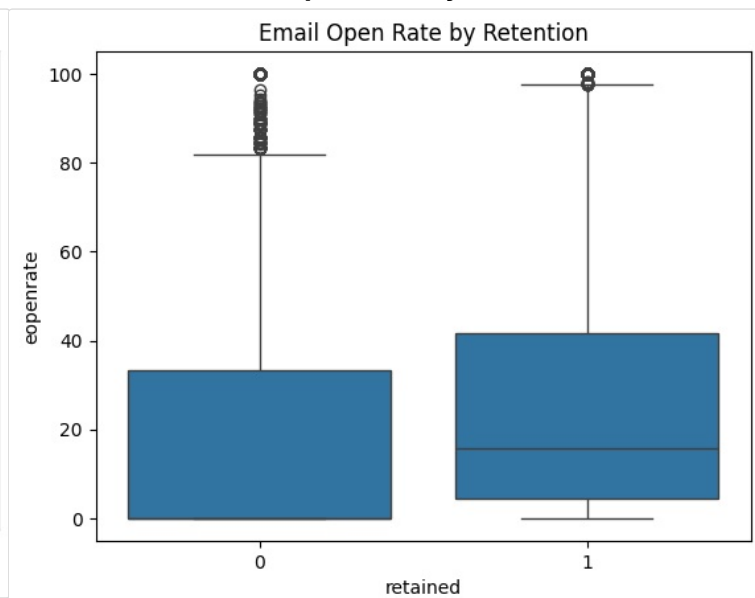
Retained customers have a higher average order value.

3. Average Order Frequency by Retention



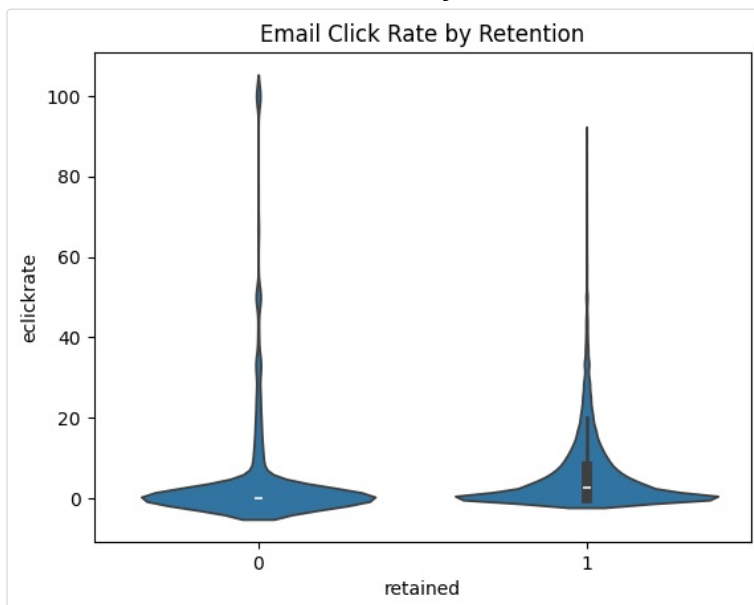
Retained customers usually have a higher order frequency.

4. Email Open Rate by Retention



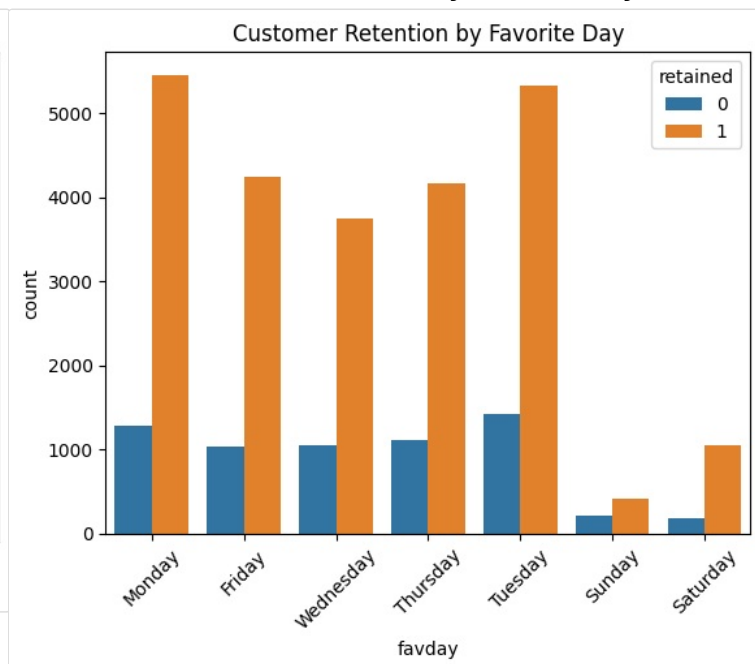
Customers who were not retained had a higher email open rate.

5. Email Click Rate by Retention



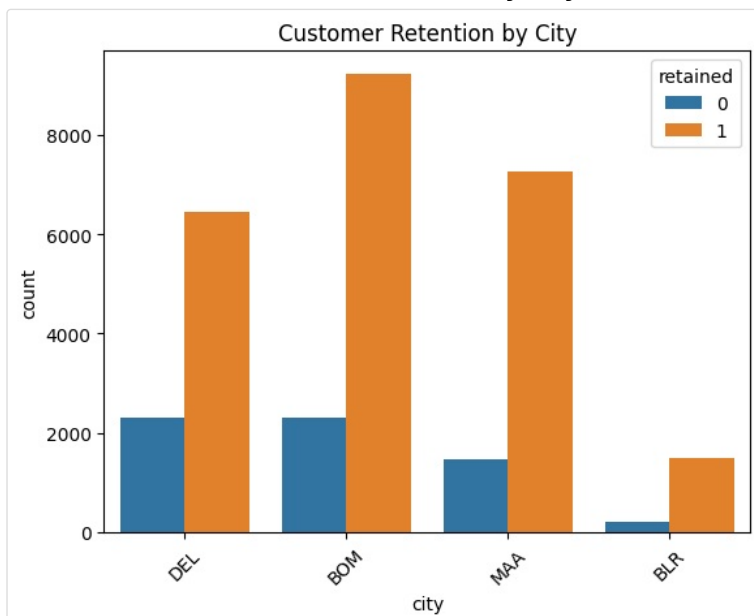
Retained customers usually have a higher email click rate.

6. Customer Retention by Favorite Day



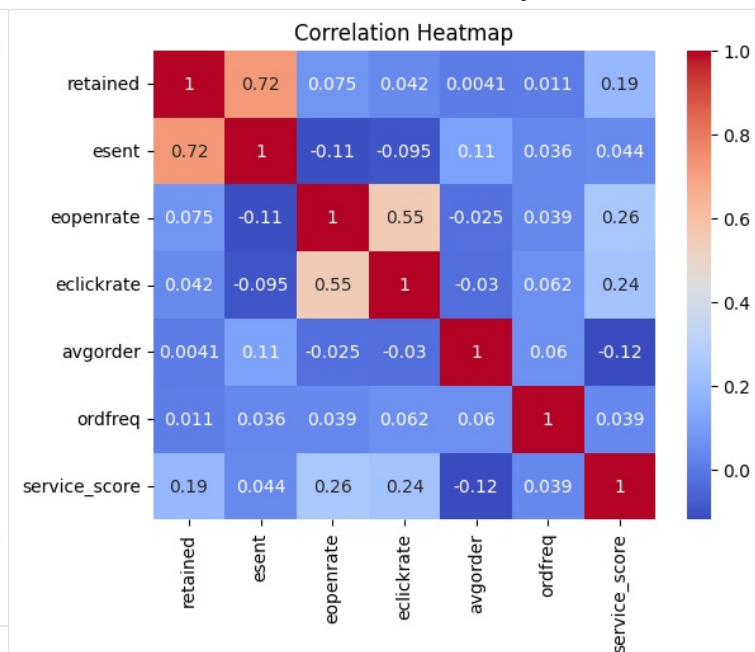
Monday had the highest number of retained customers, followed by Tuesday.

7. Customer Retention by City



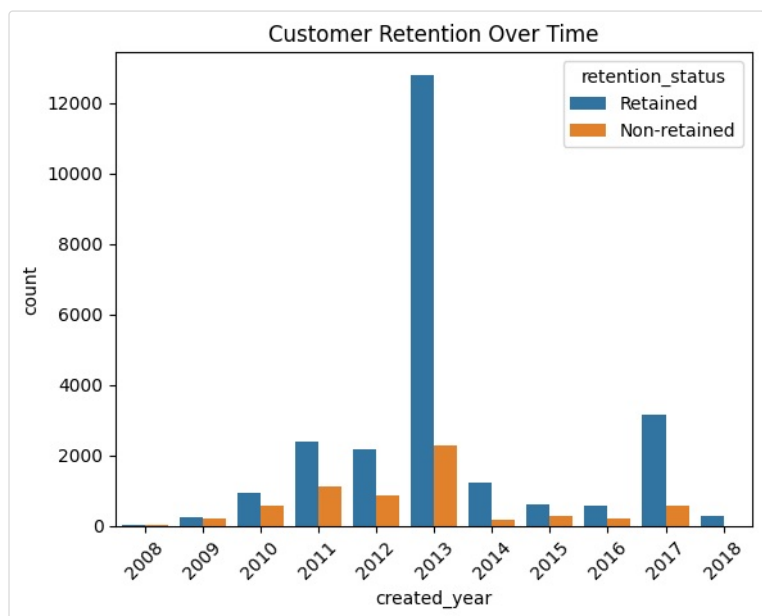
BOM city has the highest number of both retained and non-retained customers.

8. Correlation Heatmap



Email sent (esent) has the strongest relationship with customer retention.

9. Customer Retention Over Time



Customer retention was lower in 2008 and highest in 2013.

5. Conclusion:

Q1: What percentage of customers were retained?

About 79.46% of customers were retained in the dataset.

Q2: How does order behaviour differ between retained and non-retained customers?

Retained customers had a higher average order value and a higher order frequency compared to non-retained customers.

Q3: How does email engagement differ between retained and non-retained customers?

Non-retained customers had a higher email open rate, while retained customers had a higher email click rate.

Q4: Which day and city had the highest number of retained customers?

Monday had the highest number of retained customers, followed by Tuesday, and BOM city had the highest number of both retained and non-retained customers.

Q5: Which feature had the strongest correlation with retention, and in which years was retention lowest and highest?

esent (number of emails sent) had the strongest correlation with retention. Retention was lowest in 2008 and highest in 2013.